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A paleomagnetic investigation of the Minturn Formation, Colorado: a study in establishing the timing of remanence acquisition

Greg Magnus and Neil D. Opdyke
Department of Geology, University of Florida, Gainesville, FL 32611, USA

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ABSTRACT

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One hundred sixty-five oriented hand samples were collected from 55 sites in the Minturn Formation located between Wellsville and Howard, Colorado, in an attempt to locate the base of the Permo-Carboniferous Reversed Superchron (PCRS) in western North America. After the samples were demagnetized using thermal and alternating field cleaning techniques, the normal polarity magnetizations (which marks the base of the Permo-Carboniferous Reversed Superchron (PCRS)) were not found in the Minturn Formation. Reversed polarities were consistent throughout the section studied.

A conglomerate test was conducted on syndepositional rip-up clasts taken from the formation. The test indicates that a post-depositional, secondary component of magnetization has been acquired by the clasts and remains stable up to temperatures of 610–640°C. However, the high temperature components for three of the four test sites show random directions indicating that a primary remanent magnetization was acquired shortly after deposition. The conglomerate test demonstrates that a secondary Permian direction can be successfully isolated from a primary Pennsylvanian one revealed at blocking temperatures above 650°C. In this study, the two components were isolated using step-wise thermal demagnetization.

The results of the study indicate that the base of the PCR superchron lies stratigraphically below the Minturn Formation, probably in Early Atokan or Morrowan age sediments.

Introduction

The extension of the magnetic polarity time scale into the Paleozoic has shown that there exists a potentially important time-stratigraphic horizon of global significance within the Carboniferous. The horizon is known as the Permo-Carboniferous Reversed Superchron (PCRS), or the Kiaman Superchron (Irving and Parry, 1963). During this interval (Mid Carboniferous to Late Permian), the polarity of the Earth's magnetic field was dominantly reversed for approximately 50 m.y. making the normal polarity zone at the base of the interval an important magnetostratigraphic marker for world-wide correlations. Previous work in western North America by Miller and Opdyke (1985) had indicated that the base of the interval is located below the upper part of the Minturn Formation of Colorado.

Description and geological setting of area investigated

Outcrops of the Minturn Formation were sampled in the Arkansas River Valley and its tributaries between Wellsville, and Howard, Fremont County, Colorado (Fig. 1). Approximately 1280 m of Minturn strata are exposed at this locality (Brill, 1952). The base of the Minturn conformably overlies shale and sandstone correlated with the Pennsylvanian Belden Formation. The top of the Minturn is conformably overlain by the Pennsylvanian and Permian Sangre de Cristo Formation. All of the strata of the Minturn Formation

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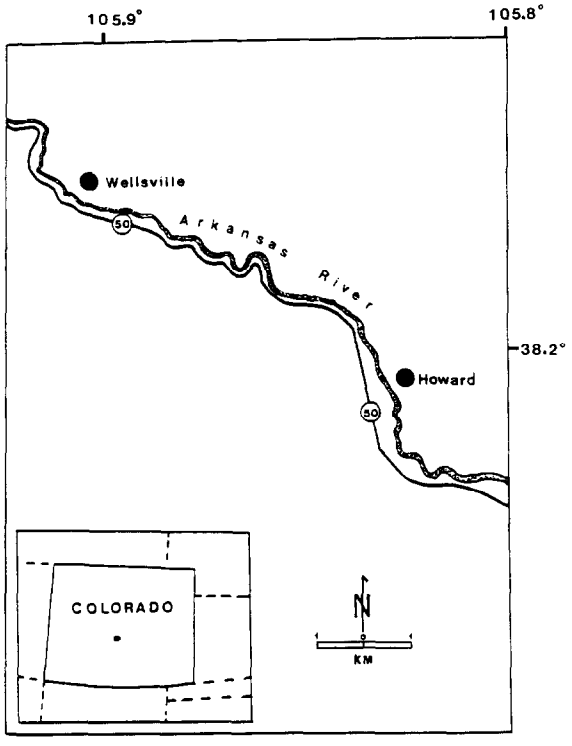


Fig. 1. Locality of the Arkansas River Sections.

are regarded as Middle Pennsylvanian, Atokan–Demoinesian in age (Tweto and Lovering, 1977). The formation was deposited between the Front Range and Uncompaghre Highlands, which extended through northern New Mexico and Colorado during the Carboniferous. These two relatively large highlands contributed sediment to a rapidly subsiding basin, resulting in the huge sediment accumulations which created the Minturn Formation. The Minturn strata, red coarse clastic units that consist of intertonguing marine and non-marine deposits, are interpreted as a series of stacked fan-delta deposits that formed in this central Colorado trough (Lindsey et al., 1985).

Sample treatment

One hundred and sixty-five oriented hand samples were collected from the Minturn Formation between Wellsville and Howard, Colorado. The samples were returned to the laboratory and subsequently drilled with a standard drill press to obtain cores of 2.4 cm diameter. From the cores, specimens of 2.5 cm length were cut for paleomagnetic study. An additional 32 core samples were collected from the base of the formation using a gasoline-powered, water-cooled, hand drill. Before removal from the outcrop, the cores were oriented with a Brunton compass and inclinometer using standard procedures (Tarling, 1983). Prior to treatment, all of the samples were stored in an unshielded room. Both thermal and alternating-field cleaning techniques were employed to demagnetize the samples.

The samples were thermally demagnetized in a Schonstedt TSD-1 Thermal Demagnetizer with a heating chamber DC field of 1×10^{-7} T and a cooling chamber DC field of less than 5×10^{-9} T. After measurement of the NRM directions which are directed toward the southeast with shallow inclinations, a pilot set of samples representing the sample set were heated for approximately 40 min at each of the following temperature increments: 100, 200, 300, 400, 450, 500, 525, 550, 575, 600, 610, 620, 630, 645, 660, 670 and 680°C. After each of the above treaments, the sample directions and intensitites were measured using a SCT cryogenic Magnetometer.

Alternating-field demagnetization was accomplished using a GSD-1, single-axis, A.C. geophysical specimen demagnetizer. This system reaches peak fields of 100 mT.

Demagnetization results

Peak alternating fields of up to 100 mT did not significantly decrease the magnetic intensity or change the magnetic direction of the red siltstone samples analyzed in this study. This result is consistent with the fact that the magnetization in red sediments is often carried by single domain hematite which has a high coerciviety (Turner, 1980). Therefore, thermal demagnetion was employed to magnetically clean the samples.

The majority of orthogonal plots (Fig. 2; Zijderveld, 1967) show what appears to be the consistent decay to the origin, however, in most samples a change in declination and inclination is observed at high temperatures. The two components isolated have similar directions, with the low-temperature component being unblocked between 100–640°C. Principle component analysis

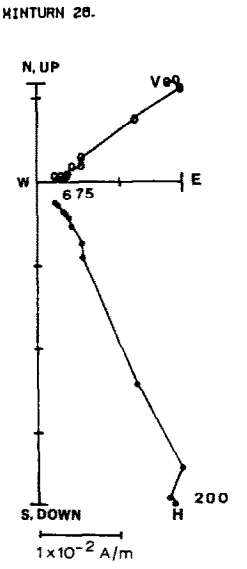


Fig. 2. Orthogonal plots of horizontal (filled circles) and vertical (open circles) components of magnetization with increasing temperature.

for both components was performed following Kirschvink, 1980. Site mean directions were calculated from samples with maximum angular deviations (MAD) of less than 10°. In situ and tilt corrected site mean directions with α_{95} values of less than 25 are plotted in Fig. 3, for the high temperature components. The overall formation means were calculated for both components following Fisher (1953). The low temperature component has a mean declination of 140.1 and inclination of -8.2 ($\alpha_{95} = 3.4$, $k = 15.3$, $R = 112$, and $N = 120$). The characteristic high temperature component is isolated between 640°C and 680°C, and has a mean declination of 136.2° and inclination of $-4.3°$ ($\alpha_{95} = 4.9$, $k = 24.1$, $R = 35.5$, and $N = 37$).

Thirty-two samples of both red siltstones and limestones were collected for study from the lower 50 m of the Minturn Formation. The samples were prepared and treated as previously discussed. Throughout thermal cleaning, the remnent magnetizations in the red siltstones maintained reversed polarities similar to the rest of the formation. The directions from seven limestone samples yielded Zijderveld diagrams with no internal consistency (i.e., no linear components could be defined statistically using the principal component analysis of Kirschvink (1980)]. The rest of the limestone samples from the base of the section yielded high temperature directions similar to the red siltstones sampled through the formation.

Conglomerate test

A conglomerate test (Graham, 1949) was conducted on syn-depositional rip-up clasts taken from four different sites within the Minturn Formation in an attempt to ascertain the acquisition time of remanent magnetization, one of the most perplexing problems in red-bed paleomagnetism. Both clast and matrix samples were collected in the field using a gasoline-powered, water-cooled,

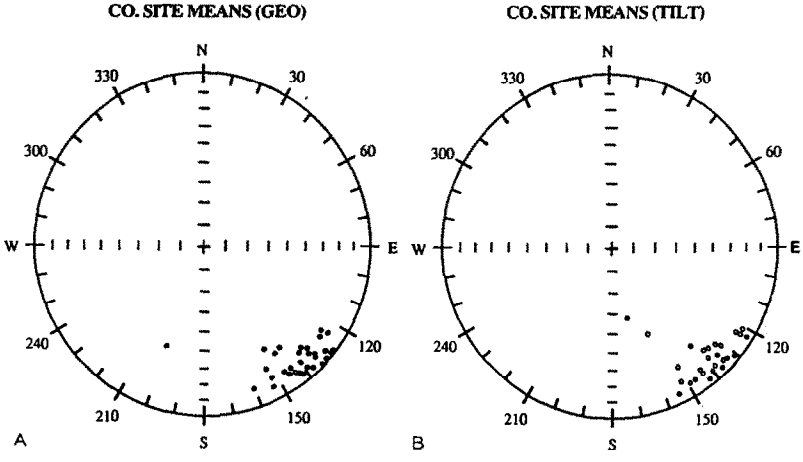


Fig. 3. Site mean directions for high temperature components plotted on an equal angle stereographic net. The open circles (filled circles) indicate directions on upper (lower) hemisphere before (A) and after tilt correction (B).

hand drill. The samples were oriented before removal from the outcrop with a Brunton compass and inclinometer. Subsequent to drilling, the samples were taken back to the laboratory where the clasts were meticulously separated from any matrix using a standard rock cutting saw. Extreme care was taken to maintain the integrity of the sample orientations throughout this process. Samples were small, usually less than 5 cm^3 and having irregular shapes.

Upon completion of sample preparation, the clast and matrix specimens were thermally demagnetized using the equipment discussed previously. Each sample was subjected to stepwise thermal demagnetization up to a temperature of 680°C.

Subsequent to thermal treatment, orthogonal projections were plotted for the clast (Fig. 4) and matrix samples (Fig. 5). The orthogonal diagrams show two component behavior for the majority of the clasts. A low temperature (100–620°C) component directed towards the southeast with low inclination and a high temperature component above 620°C which is randomly directed. The low temperature directions observed from the clasts are similar to the low temperature directions observed for samples throughout the section, and show a cluster of directions in the southeast quadrant with shallow negative inclinations (Fig. 6). The mean low temperature direction derived from the clasts is $D = 147.8$ and $I = 11.5$ ($\alpha_{95} =$

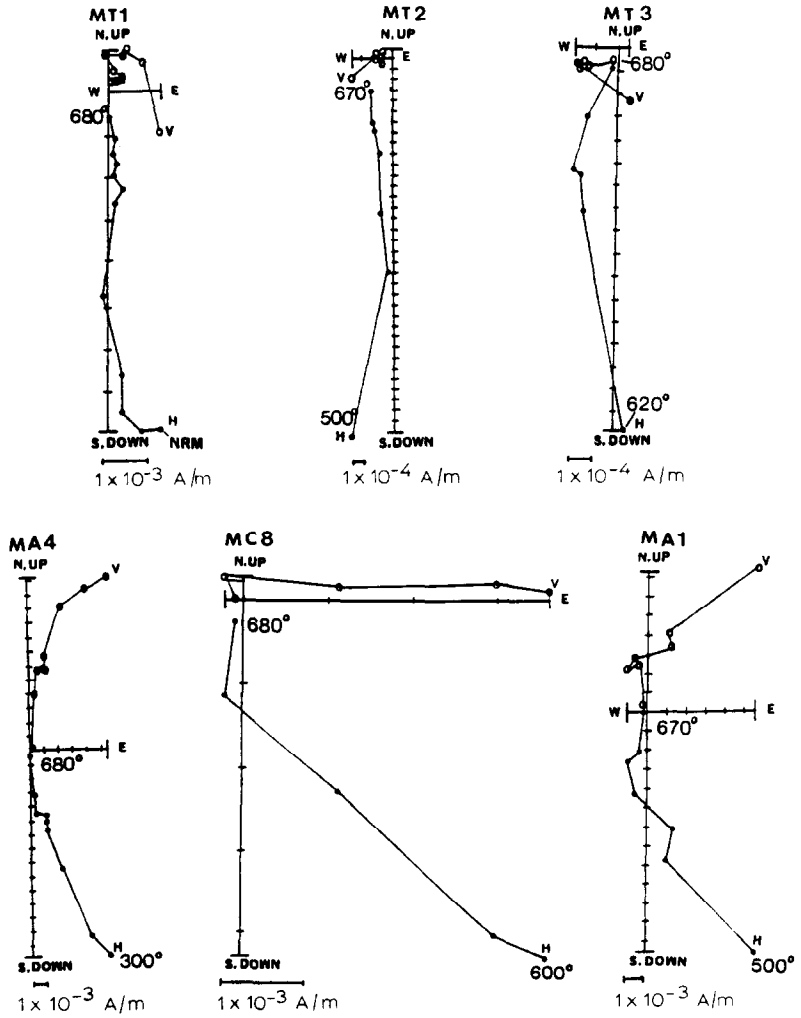


Fig. 4. Orthogonal projections of the directions of magnetization obtained from the clasts. Symbols the same as in Fig. 2.

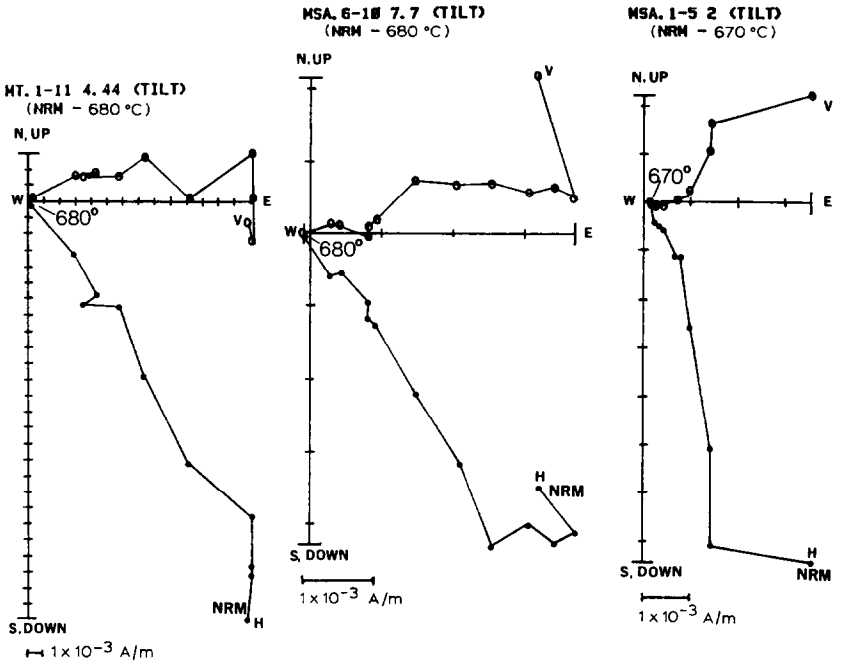


Fig. 5. Orthogonal projections of the directions of magnetization from the coarse sandstone matrix from which the clasts were obtained. Symbols are the same as in Fig. 2.

14.1, $N = 20$, $k = 6.3$, $R = 17$). This direction is similar to that obtained for the low-temperature component from the entire formation (Table 1) and is probably of Early Permian age.

High temperature directions were determined from the clasts using principal component analysis. These directions are shown in Fig. 7. The results from the four sites studies were analyzed according to the test of randomness proposed by Watson (1956). The high temperature components from site MS are not randomly magnetized. They do, however, give a direction different from that seen throughout the formation. However, the R values (resultant vector obtained from N samples) for sites MD, MA and MT do not exceed the significance values given by Watson and are,

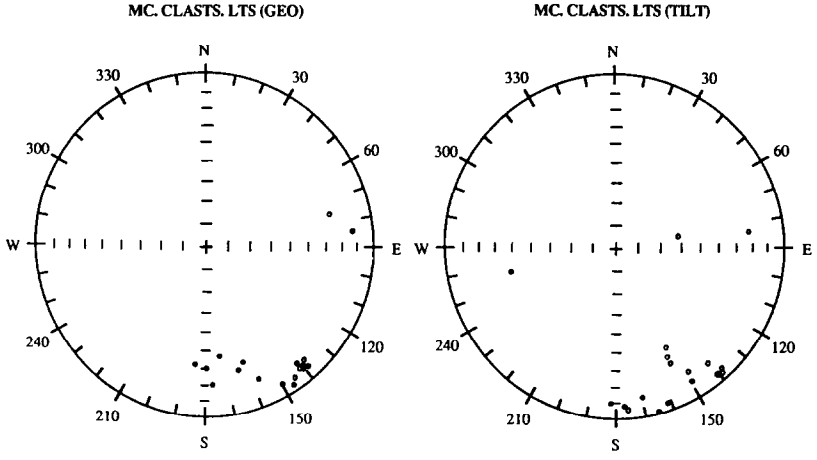


Fig. 6. Low temperature component, from clasts of the Minturn Formation corrected for tilt. Symbols as in Fig. 3.

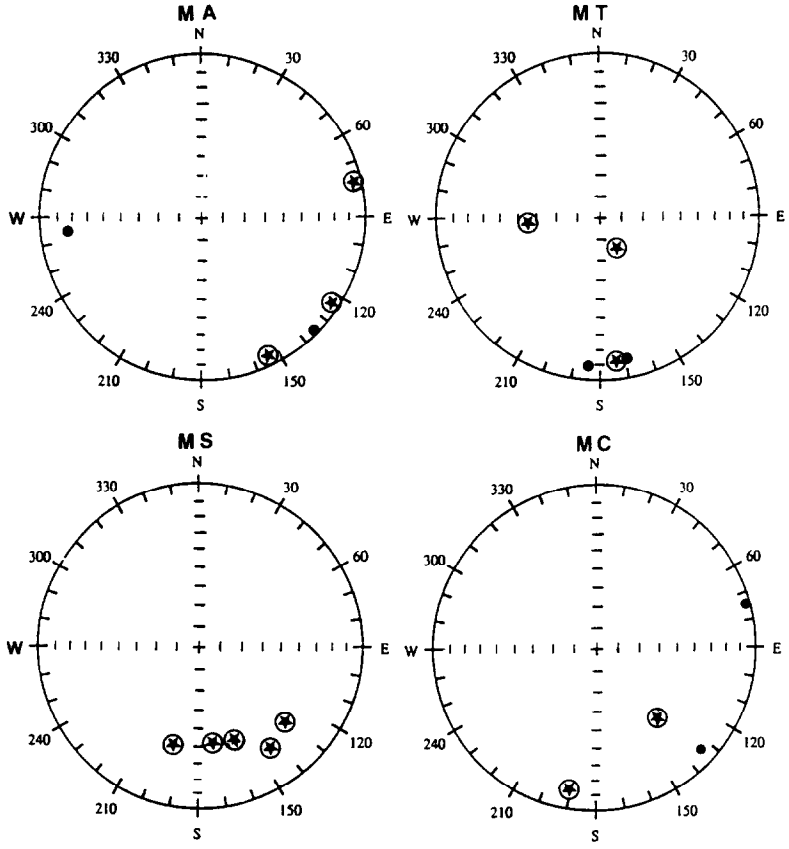


Fig. 7. High temperature components from four sites containing clasts. Stars represent directions on the upper hemisphere while filled circles represent direction on the lower hemisphere.

therefore, representative of random populations (see Table 2). The results indicate that three of the sites MT, MA and MC possess random components isolated at temperatures between 640°C and 680°C (Fig. 7) indicating that the characteristic remanent magnetization has been acquired by

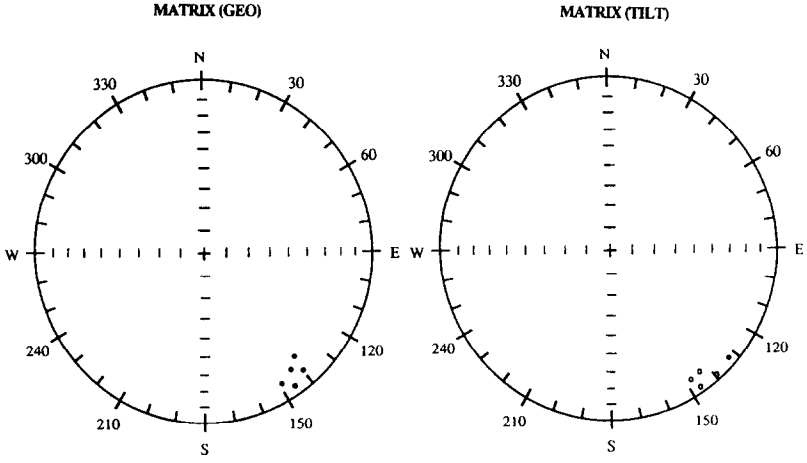


Fig. 8. Directions of magnetization from the Coarse Sandstone Matrix. Symbols are the same as in Fig. 3.

TABLE 1

Comglomerate test sites *

Site	N	R	R_0	Note
MA	5	2.84	3.50	Pass
MC	4	2.66	3.10	Pass
MS	5	4.76	3.50	Fail
MT	5	3.30	3.85	Pass

* N = number of samples, R = resultant vector of samples, R_0 = significance point of Watson (1956).

a majority of the clasts at or shortly after the samples were deposited. These results are similar to findings obtained from Miocene red clasts from the Siwalik Molasse of Pakistan (Tauxe and Opdyke, 1982).

In comparison to the clast samples, the samples from the matrix show a well grouped cluster of directions in the southeast quadrant (Fig. 8), and it can be observed that these directions are similar to expected Permian directions.

Conclusion

From the data collected in this study, it can be concluded that the base of the Permo-Carboniferous Reversed Interval does not lie within the Minturn Formation at the Arkansas River Valley section of Colorado. The formation contains exclusively reversed remanent magnetizations from top to bottom, with a formation mean direction of declination 136.2° and inclination −4.3° (α_{95} = 4.9, k = 24.1, R_{35} = 5, N = 37 sites, consisting of 111 samples.

The conglomerate test indicates that the formation acquired its high temperature magnetic remanence shortly after deposition. The test also indicates that a pervasive low unblocking temperature component of Permian age has been acquired by the entire formation. The secondry post-depositional remanence was removed thermally and a characteristic remanence isolated at temperatures above 620–640°C. The use of the conglomerate

TABLE 2

Pole positions calculated from this study and other North American Carboniferous pole positions

Formation	Symbol	Ded. (°)	Ind. (°)	Paleopole lat. (°)	Paleopole long. (°)	α_{95} (°)	K	R	N
Minturn Fm.	Mh	136.2	−4.3	36.6	133.1	4.9	24.1	35.5	38 sites
Minturn Fm. (low temp.)	Ml	140.1	−8.2	141.0	129	3.4	15	11	120 samples
Clasts (low temp.)	Mc	147.8	−11.5	46.2	124.6	14.0	6.3	17.0	20 samples

Other North American Late Paleozoic poles

Formation	Age	Symbol	Paleopole long. (°)	Paleopole lat. (°)	α_{95} (°)	Source
Prince Ed Is	Early Permian	Pe	125.5	42.1		Symons (1990)
Joggins	Westphalian B	Ja	113.0	41.8	4.8	DiVenere and Opdyke (1991)
Joggins	Westphalian B	Ju	120.3	37.2	4.7	DiVenere and Opdyke (1991)
Boss Point	Westphalian A	BP	120.6	39.6	8.3	DiVenere and Opdyke (1991)
Claremont	Westphalian A	C	123.6	39.4	5.8	DiVenere and Opdyke (1991)
Shepody	Namurian	Sh	124.3	35.7	4.6	DiVenere and Opdyke (1990)
Maringouin	Namurian A	Ma	121.8	32.4	3.8	DiVenere and Opdyke (1990)
Minturn and Maroon	Permo-Carboniferous	MM	115.4	42.6	2.8	Miller and Opdyke (1985)

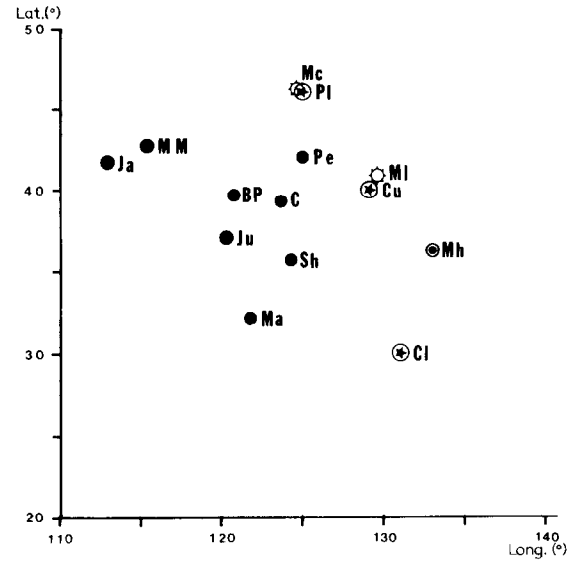


Fig. 9. Carboniferous and Permian pole positions for North America. Mean poles for Early Permian (*Pl*), Early Carboniferous (*Cu*), and Middle Carboniferous (*Cl*) are indicated by stars and are taken from Van der Voo (1990). *Mh* (circled point) = the pole position from the high temperature component of the Minturn Formation; *Mc* = from the low temperature component of the clasts, *Ml* = low temperature from the Minturn Formation and are indicated by an anulis. Other poles are identified in Table 2 and are indicated by the filled circles.

test has enabled us to distinguish between two similar but significantly different vectors.

Pole positions were calculated using the method of statistical analysis given by Fisher (1953), and are plotted against the polar wander path of Van der Voo (1990), as well as several Carboniferous paleopole positions published since the appearance of Van der Voo's paper. These are listed in Table 2, see Fig. 9.

The pole from the Minturn formation, derived from its high temperature component, lies between the mean Late Carboniferous and Middle Carboniferous paleopoles (Van der Voo, 1990) without any significant offset from the polar wander trend. The pole also lies close to the pole position published by DiVenere and Opdyke (1990) from the restudy of the Shepody formation of New Brunswick and also close to paleopoles from the Cumberland Group (DiVenere and Opdyke, 1991).

It is interesting to compare the pole position from the Minturn and Maroon formations at Vail, Colorado (Miller and Opdyke, 1985), with that presented here, it can be seen that the latitude of the pole position is similar but that the longitudes differ by approximately 18°. This may indicate that a structural rotation has taken place between the two sites. However, the Vail data are younger than that from the Arkansas River section since it represents results from the Early Permian Maroon Formation, as well as from the uppermost Minturn Formation.

It is significant that the low temperature pole position determined from the formation lies along the Early Paleozoic trend of the pole positions and is almost coincident with the average Late Carboniferous paleopole of Van der Voo (1990). It is, therefore, probable that the overprint is of Late Carboniferous or Early Permian age. It is not significantly different from the low temperature component from the clasts. This direction in the absence of the conglomerate test would usually be taken as the characteristic remanent direction for the formation. In this case, it can be seen as resulting from two components, one Permain and one Pennsylvanian thereby, resulting in a pole position which is precise but in error by some undetermined amount. In light of this study, it is clear that many of the pole positions which are used from prior studies may in fact be a resultant of two components which are very difficult to separate. Paleopoles calculated from this data will clearly determine the polar wandering trend but will contain indeterminate amounts of error due to the overprint. Many of these pole positions could probably be improved by re-examination.

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